



Kotlin

Training Assignment

Document Code	25e-BM/HR/HDCV/FSOFT
Version	1.0
Effective Date	31/08/2023

RECORD OF CHANGES

No	Effective Date	Change Description	Reason	Reviewer	Approver

Contents

Assignment 01: Coroutines Basic	4
Assignment 02	4
Assignment 03. Coroutines Advance	5



CODE	:	KOL_Assignment05_v1.0
TYPE	:	N/A
LOC	:	N/A
DURATION	:	180 Minutes

Assignment 01: Coroutines Basic

Build a coroutine that calculate the Nth Fibonacci number, then return the result as an Integer .

Technical Requirements:

- CoroutineScope, Dispatchers, runBlocking, suspend function

Assignment 02

Build a parent coroutine with three sub – coroutine inside it, each sub coroutine return different kinds of exception, using to validate username. The parent coroutine will take an input as a string, and return a Boolean result (Yes is valid). A general rule for username is that the length should be at max 16 character, and do not contain Upper case characters. The logic of filtering is described as below:

Sub Coroutine 1 will examine if the username contains any number, if have then it should throw a custom exception called InvalidNumberException.

Sub Coroutine 2 will examine if the string contain special characters which are !@#\$%^&*()_, if have then it should throw a custom exception called InvalidSpecialCharacterException.

Sub Coroutine 3 will examine if the length of string is longer than 4 or not, if not then it should return InvalidLengthException.

For other rules, if anything is break then the parent coroutine should return a. generic exception called InvalidUsernameException.

Technical Requirements:

- CoroutineScope, Dispatchers, runBlocking, suspend function

Assignment 03. Coroutines Advance

Build a function that return sum of integer number from 0 to N, then build a flow that return the result of this function for input parameter from 0 to 10. Flow should return the result 10 times, 500ms is the time delayed between each time.

Technical Requirements:

- Flow operator, delay

-- THE END --